

Additional text 6: does reading several ESMC layers at once predict better?

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Question

The best single ESMC layer differs between responses: layer 50 for salt and staurosporine, the final layer (80) for heat and the DNAJB11-versus-DNAJA1 preference. Does combining layers predict better than the best single layer, and which layers does each response draw on?

Method

Nine mean-pooled hidden states (layers 0, 10, $\dots$, 80) for every protein, eight responses, the usual outer folds grouped by sequence cluster. Within each training fold, a second, inner grouped split makes every choice, so the outer test fold is never used to choose anything:

- **each layer alone:** ridge regression, penalty chosen on the inner split;
- **chosen single layer:** the layer that scores best on the inner split, a fair stand-in for “the best layer”, which would otherwise be picked by looking at the test results;
- **stacking:** the nine single-layer models combined with non-negative weights fitted to their inner out-of-fold predictions;
- **concatenation:** all nine layers in one ridge model, each layer standardised and scaled so that each contributes equally.

Penalties come from a grid of seven values between 300 and 300,000; the staurosporine responses are sensitive to this choice (their R2 ranges from -20 % to +6 % across the grid), so a coarse grid understates them. Intervals resample sequence clusters. 2.1 minutes on one RTX card for all eight responses.

Results

Response	L0	L10	L20	L30	L40	L50	L60	L70	L80	Chosen	Stacked	Concat.
Salt 75 mM	6.2	13.2	13.9	14.7	14.5	16.0	14.7	14.8	15.8	16.0	18.0	17.8
Salt 150 mM	7.4	14.8	16.0	16.6	16.3	18.2	16.2	17.0	18.1	17.7	20.4	19.5
Heat 43 °C	8.9	20.2	20.7	21.1	20.9	22.5	20.8	21.1	25.7	25.7	27.4	26.4
Heat 37 °C	3.4	9.3	9.4	9.4	9.6	10.7	10.2	10.2	11.4	11.4	12.7	12.3
Stau. 35 °C	1.6	5.0	6.1	6.4	6.3	6.0	4.9	4.6	5.0	5.9	6.6	6.6
Stau. 37 °C	0.4	5.8	6.2	6.1	6.1	5.9	6.5	5.8	4.3	5.7	6.6	6.5
Stau. 43 °C	0.6	2.8	3.1	3.8	3.7	4.1	3.2	2.9	2.0	2.4	3.5	3.7
DNAJB11 pref.	30.6	45.2	46.0	46.2	45.7	43.5	36.2	38.6	45.6	46.1	49.1	48.6

Percent of variance explained on held-out sequence clusters.

Response	Stacked minus chosen layer (95 % interval)	Layers stacking relies on (mean weight)
Salt 75 mM	+2.0 [+1.3, +2.7]	80 (0.35), 50 (0.24), 60 (0.20)
Salt 150 mM	+2.6 [+1.8, +3.5]	80 (0.35), 50 (0.23), 60 (0.17)
Heat 43 °C	+1.6 [+1.0, +2.2]	80 (0.59), 60 (0.15), 10 (0.14)
Heat 37 °C	+1.2 [+0.7, +1.7]	80 (0.51), 50 (0.21), 60 (0.17)
Stau. 35 °C	+0.7 [+0.1, +1.3]	20 (0.39), 80 (0.21), 60 (0.16)
Stau. 37 °C	+1.0 [+0.5, +1.4]	60 (0.34), 20 (0.31), 80 (0.19)
Stau. 43 °C	+1.1 [+0.6, +1.8]	60 (0.59), 20 (0.23), 0 (0.13)
DNAJB11 pref.	+3.0 [+2.3, +3.6]	80 (0.36), 40 (0.22), 10 (0.17)

Conclusions

- **Combining layers helps every response.** Stacking beats the chosen single layer on all eight, by 0.7 to 3.0 points, with every interval above zero; concatenation does about as well (within 1 point of stacking). The best figures are now 49.1 % for the co-chaperone preference, 27.4 % for heat at 43 °C and 20.4 % for salt at 150 mM.
- **Different responses read different depths.** Heat draws mostly on the final layer; salt on the final layer together with layers 50-60; staurosporine on layer 20 and layer 60; the co-chaperone preference on layers spread across the network (80, 40, 10).
- **Why the sparse autoencoder lost on the preference.** For the DNAJB11 preference, layer 60 is the weakest of the middle layers (36.2 %, against 45-46 % at layers 10-40 and 80). The sparse autoencoder reads layer 60, which is why its features explained 7 points less of the preference than the embedding did, while matching it for the other responses.
- **Size of the gain.** One to three points of variance is a 6.5-15 % relative improvement for salt, heat and the preference (largest for salt at 150 mM, 17.7 to 20.4 %). For staurosporine, the absolute gain is under a point.

Critical evaluation

- **Penalty sensitivity.** With a coarser grid the staurosporine single layers scored 1-2 points lower and stacking appeared to gain 1.2-1.6 points; the finer grid is used here, and the staurosporine gains shrink to 0.7-1.1. Choosing the penalty on the inner split remains noisier for staurosporine than for the other responses, so its single-layer figures still sit up to 1.6 points below a fit whose penalty was chosen on the test folds (the optimistic figure used in some earlier tables).
- **Nine of 81 layers.** Only every tenth hidden state was pooled; finer sampling could move the best single layer and the stacking weights slightly.
- **Linear readouts only.** Stacking was not combined with the neural-network readouts, which gained a similar 1-2 points on heat and salt; whether the two gains add is untested.
- **The weights are descriptive.** They show which layers carry non-redundant information under ridge, not where in the network a property is computed.

Methods files

code/ad06_layers.py, slurm/ad06_layers.sbatch; results in data/ga_data/ad06_results.json.